

Practical 6 - Competing risks 1

Survival Analysis

London School of Hygiene & Tropical Medicine

Stata version (please see accompanying Stata do file)

SOLUTIONS

Datasets required: `aaatrial.dta`

Introduction

In this practical we will use a dataset from a randomised trial of screening for abdominal aortic aneurysms (AAA), in which older men were randomised to be invited to be screened for AAA (invited group) or not (control group). The main outcome of interest was deaths relating to AAA (including deaths following AAA repair operations), but men in the study also died from other causes. There are 13,562 people in the data.

If an AAA was found at screening, a repair operation was carried out, essentially removing the risk of future death from AAA. However, the operation itself carries around a 5% risk of mortality. Undetected AAAs may rupture at any time, resulting in either an emergency repair operation (which carries around a 40% risk of mortality) or death. Participants were recruited to the trial between 1997 and 1999 and followed up until the end of 2007.

The variables used in this practical are described below.

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>group</code>	Randomisation group: 0=Control, 1=Invited
<code>dateran</code>	Date of randomisation
<code>aaadeath</code>	AAA-death indicator: 0=Censored or non-AAA-death, 1=AAA-death
<code>allddeath</code>	All-cause mortality: 0=Censored, 1=Death (any cause)

Variable	Description
deathtype	0=Censored; 1=non-AAA death; 2=AAA-death
timeout	Date of study exit

Aims

The aims of this session are:

- Estimate cumulative incidence functions using non-parametric methods (the Aalen-Johansen estimator), and plot and interpret these.
- Compare cumulative incidence functions between groups.
- Understand why the Kaplan-Meier estimator is not in general suitable for estimating cumulative incidences when there are competing events.

Question 1

- Load the data.
- How many people died from any cause, and how many had an AAA-death and a non-AAA-death?

SOLUTION

There are 4040 deaths. Of these, 105 are AAA-deaths and 3935 are deaths from other causes (non-AAA-deaths).

Question 2

We will begin by focusing on death from any cause using the `alldeath` variable.

- Use the following `stset` code:

```
stset timeout, fail(alldeath) origin(dateran) id(id) scale(365.25)
```

- Plot the Kaplan-Meier survival curves for all-cause mortality.
- What is the risk of death from any cause up to 5 years?

SOLUTION

The survival probability at 5 years is 0.86. The risk of death from any cause up to 5 years is therefore 14%.

Question 3

We will now move on to considering AAA-deaths and non-AAA-deaths as competing events. In Stata re-stset your data using the `deathtype` failure variable as shown below, indicating AAA-death as failure. Study the output from `stset`}‘ and how it differs from that in question 2.

Take a look at `_t` and `_d` for the first three individuals in the data. What information is given for (i) people who have an AAA-death, (ii) people who have a non-AAA-death, (iii) people who are censored?

```
stset timeout, origin(dateran) id(id) fail(deathtype==2) scale(365.25)
```

SOLUTION

- The first person in the data died from an AAA-death at 3.78 years and has `_d=1`. The second person was censored at 10.32 years and has `_d=1`. The third person died from a non-AAA-death at 0.219 years and has `_d=0`, i.e. as if they were censored. This is OK, it is just how Stata stores the information. Below we will tell Stata how to treat non-AAA-deaths when our interest lies in the cumulative incidences of AAA-deaths, and vice-versa.

Question 4

- Estimate the cumulative incidences for AAA-deaths and non-AAA-deaths using the following code

```
**You may need to install the stcompet function like this
ssc install stcompet

** Generate the cumulative incidences for each event type
stcompet cuminc=ci, compet1(1)

gen cuminc_AAA = cuminc if deathtype==2
gen cuminc_nonAAA = cuminc if deathtype==1
```

- The resulting cumulative incidence curves can be plotted using

```
twoway (line cuminc_AAA _t if deathtype==2, connect(step) sort lcolor(black)) ///
(line cuminc_nonAAA _t if deathtype==1, connect(step) sort lcolor(black) lpattern(dash)) ///
, legend(lab(1 "non-AAA deaths") lab(2 "AAA deaths"))
```

- What are the cumulative incidences for AAA-deaths and non-AAA-deaths up to 5 years?

SOLUTION

The cumulative incidence of AAA-death is very low, and much lower than the cumulative incidence of non-AAA-death. The estimated cumulative incidence of AAA-death up to 5 years is 0.79%. The estimated cumulative incidence of non-AAA-death up to 5 years is 29.3%.

Question 5

- Returning to considering death from any cause (`alldeath`), obtain Kaplan-Meier plots by treatment group.
- What is the risk of death from any cause up to 5 years in the two groups?
- Use a log-rank test to test the null hypothesis that there is no difference between the survival curves in the two groups.

SOLUTION

- There is very little difference between the overall survival curves in the two treatment groups. The estimated survival probability at 5 years is 0.859 in the control group and 0.861 in the invited group.
- The log rank test has a p-value of 0.3. There is no evidence against the null hypothesis of no difference between the survival curves in the two groups.

Question 6

We now return to considering the competing events of AAA-death and non-AAA death.

- Estimate the cumulative incidence function for AAA-deaths and non-AAA-deaths in each treatment group by modifying the code used in question 4 - obtain the cumulative incidences using `stcompet` separately by group.
- What are the cumulative incidences for AAA-deaths and non-AAA-deaths up to 5 years in the two groups?
- A test of the null hypothesis of no difference in the cumulative incidences for a given cause of death by group can be performed as follows:

```
stpepemori group, compet(1)
```

SOLUTION

- The estimated cumulative incidence of AAA-death up to 5 years is 0.96% in the control group and 0.61% in the invited group. The estimated cumulative incidence of non-AAA-death up to 5 years is 29.7% in the control group and 29.0% in the invited group.

- The test of Pepe and Mori gives no evidence against the null hypothesis of no difference between the cumulative incidences (for either event type) in the two groups. In this example we get different results from this test compared to Gray's test, which does not appear to be available in Stata.
- Gray's test (not done in Stata) gives a p-value of 0.498 for non-AAA-deaths, indicating no evidence against the null hypothesis of no difference between the cumulative incidences in the two groups.
- Gray's test gives a p-value of 0.043 for AAA-deaths, indicating some evidence against the null hypothesis of no difference between the cumulative incidences in the two groups.

Question 7

- Use the Kaplan-Meier estimator to estimate the cumulative incidence of AAA-death using

```
stset timeout, origin(dateran) id(id) fail(deathtype==2) scale(365.25)
sts graph, failure ylabel(minmax)
```

- Use the code in the do file to plot the cumulative incidence based on the Kaplan-Meier estimate and compare this with what you obtained in Question 4 based on the Aalen-Johansen estimator. Comment on why the results are different.

SOLUTION

As expected from theory, the Kaplan-Meier estimator overestimates the cumulative incidence of AAA death, as compared to the estimate obtained using the Aalen-Johansen estimator. This is because the Kaplan-Meier estimator treats non-AAA-deaths in the same way that it treats censoring, resulting in an implicit assumption that individuals who have had a non-AAA-death could have an AAA-death after their time of having a non-AAA-death, which does not make sense.